

# Spivak's *Calculus* (4th Edition)

Complete Definitions & Theorems  
with Examples and Applications

## Contents

## Part I

# Prologue

## Chapter 1. Basic Properties of Numbers

The entire edifice of analysis rests on twelve properties of the real numbers. These are not “proved”; they are *axioms*—the rules of the game.

### 1.1. The Field Axioms (P1–P9)

**Axiom 1.1** (P1 – Associative law for addition). For all  $a, b, c$ ,

$$a + (b + c) = (a + b) + c.$$

**Axiom 1.2** (P2 – Additive identity). There exists a number 0 such that for all  $a$ ,

$$a + 0 = 0 + a = a.$$

**Axiom 1.3** (P3 – Additive inverses). For every  $a$  there exists  $-a$  such that

$$a + (-a) = (-a) + a = 0.$$

**Axiom 1.4** (P4 – Commutative law for addition). For all  $a, b$ ,

$$a + b = b + a.$$

**Axiom 1.5** (P5 – Associative law for multiplication). For all  $a, b, c$ ,

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

**Axiom 1.6** (P6 – Multiplicative identity). There exists  $1 \neq 0$  such that for all  $a$ ,

$$a \cdot 1 = 1 \cdot a = a.$$

**Axiom 1.7** (P7 – Multiplicative inverses). For every  $a \neq 0$  there exists  $a^{-1}$  such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

**Axiom 1.8** (P8 – Commutative law for multiplication). For all  $a, b$ ,

$$a \cdot b = b \cdot a.$$

**Axiom 1.9** (P9 – Distributive law). For all  $a, b, c$ ,

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

*Remark.* Properties P1–P9 make  $(\mathbb{R}, +, \cdot)$  a *field*. They justify all routine algebra: factoring, expanding, solving linear equations, etc.

*Example.* Using P9, we prove  $a \cdot 0 = 0$ : We have  $a \cdot 0 + a \cdot 0 = a \cdot (0 + 0) = a \cdot 0$ . Adding  $-(a \cdot 0)$  to both sides gives  $a \cdot 0 = 0$ . Similarly,  $(-a) \cdot (-b) = ab$  follows from P3 and P9.

## 1.2. The Order Axioms (P10–P12)

**Axiom 1.10** (P10 – Trichotomy). For every  $a$ , exactly one of the following holds:

$$(i) \ a = 0, \quad (ii) \ a \in P, \quad (iii) \ -a \in P,$$

where  $P$  denotes the collection of *positive* numbers.

**Axiom 1.11** (P11 – Closure of  $P$  under addition). If  $a, b \in P$ , then  $a + b \in P$ .

**Axiom 1.12** (P12 – Closure of  $P$  under multiplication). If  $a, b \in P$ , then  $a \cdot b \in P$ .

**Definition 1.13** (Inequality).  $a > b$  means  $a - b \in P$ .  $a < b$  means  $b > a$ .

**Definition 1.14** (Absolute value).

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

**Theorem 1.15** (Triangle Inequality). For all  $a, b$ ,

$$|a + b| \leq |a| + |b|.$$

*When to apply.* The triangle inequality is the most-used inequality in analysis. It appears in virtually every  $\varepsilon$ - $\delta$  proof: whenever you need to bound  $|f(x) - \ell|$  by splitting it into two pieces.

*Example.* To show  $|x^2 - 9| < 1$  when  $|x - 3|$  is small: write  $|x^2 - 9| = |x - 3| |x + 3|$ . If  $|x - 3| < 1$  then  $|x + 3| \leq |x - 3| + 6 < 7$ , so  $|x^2 - 9| < 7|x - 3|$ . Choosing  $|x - 3| < \min(1, \varepsilon/7)$  gives  $|x^2 - 9| < \varepsilon$ .

## Chapter 2. Numbers of Various Sorts

**Definition 2.1** (Natural numbers).  $\mathbb{N} = \{1, 2, 3, \dots\}$ .

**Theorem 2.2** (Principle of Mathematical Induction). If  $S \subseteq \mathbb{N}$  satisfies (i)  $1 \in S$  and (ii)  $k \in S \implies k + 1 \in S$ , then  $S = \mathbb{N}$ .

*When to apply.* Use induction whenever a statement  $P(n)$  is indexed by natural numbers and the truth of  $P(k)$  helps prove  $P(k + 1)$ .

*Example.* Prove  $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ . Base:  $n = 1$ ,  $1 = 1 \cdot 2/2$ . Step: assume true for  $k$ ; then  $\sum_{i=1}^{k+1} i = \frac{k(k+1)}{2} + (k + 1) = \frac{(k+1)(k+2)}{2}$ .

**Theorem 2.3** (Well-Ordering Principle). Every nonempty subset of  $\mathbb{N}$  has a least element.

**Definition 2.4.**  $n! = 1 \cdot 2 \cdots n$  (with  $0! = 1$ ). The **binomial coefficient** is  $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ .

**Theorem 2.5** (Binomial Theorem).  $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$ .

*Example.*  $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$ . This is used later to find the derivative of  $x^n$ .

**Theorem 2.6.**  $\sqrt{2}$  is irrational.

*Remark.* This shows that P1–P12 alone cannot characterize  $\mathbb{R}$ ; they hold equally well for  $\mathbb{Q}$ , which has “holes”. A thirteenth property is needed.

## Part II

# Foundations

## Chapter 3. Functions

**Definition 3.1** (Function). A **function** is a rule  $f$  that assigns to each element  $x$  in its **domain** a unique value  $f(x)$ .

**Definition 3.2** (Operations on functions). Given functions  $f, g$ :

$$\begin{aligned} (f + g)(x) &= f(x) + g(x), & (f \cdot g)(x) &= f(x) \cdot g(x), \\ (f/g)(x) &= f(x)/g(x) \quad (g(x) \neq 0), & (f \circ g)(x) &= f(g(x)). \end{aligned}$$

**Definition 3.3.** A function  $f$  is:

- **even** if  $f(-x) = f(x)$  for all  $x$  (e.g.  $f(x) = x^2, \cos x$ ),
- **odd** if  $f(-x) = -f(x)$  for all  $x$  (e.g.  $f(x) = x^3, \sin x$ ),
- **periodic** with period  $p$  if  $f(x + p) = f(x)$  for all  $x$  (e.g.  $\sin, \cos$  with  $p = 2\pi$ ),
- a **polynomial** of degree  $n$  if  $f(x) = a_n x^n + \cdots + a_0$ ,  $a_n \neq 0$ .

## Chapter 4. Graphs

Key vocabulary: coordinate system, distance formula  $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ , slope, equations of lines, circles, ellipses, parabolas, hyperbolas.

## Chapter 5. Limits

**Definition 5.1** (Limit). We write  $\lim_{x \rightarrow a} f(x) = \ell$  if: for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that for all  $x$ ,

$$0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon.$$

*When to apply.* This is the foundation of all of calculus. Every statement about continuity, derivatives, and integrals ultimately reduces to an  $\varepsilon$ - $\delta$  argument.

*Example.*  $\lim_{x \rightarrow 3} x^2 = 9$ . Given  $\varepsilon > 0$ , we need  $|x^2 - 9| < \varepsilon$  when  $|x - 3|$  is small. Since  $|x^2 - 9| = |x - 3||x + 3|$ , and  $|x + 3| < 7$  when  $|x - 3| < 1$ , choose  $\delta = \min(1, \varepsilon/7)$ .

**Definition 5.2** (One-sided limits).  $\lim_{x \rightarrow a^+} f(x) = \ell$ : for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that  $0 < x - a < \delta \implies |f(x) - \ell| < \varepsilon$ .

$\lim_{x \rightarrow a^-} f(x) = \ell$ : for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that  $0 < a - x < \delta \implies |f(x) - \ell| < \varepsilon$ .

**Definition 5.3** (Limit at infinity).  $\lim_{x \rightarrow \infty} f(x) = \ell$ : for every  $\varepsilon > 0$  there exists  $N$  such that  $x > N \implies |f(x) - \ell| < \varepsilon$ .

**Theorem 5.4** (Uniqueness of limits). If  $\lim_{x \rightarrow a} f(x) = \ell$  and  $\lim_{x \rightarrow a} f(x) = m$ , then  $\ell = m$ .

**Theorem 5.5** (Limit laws). If  $\lim_{x \rightarrow a} f(x) = \ell$  and  $\lim_{x \rightarrow a} g(x) = m$ , then:

- (1)  $\lim_{x \rightarrow a} (f + g)(x) = \ell + m,$
- (2)  $\lim_{x \rightarrow a} (f \cdot g)(x) = \ell \cdot m,$
- (3)  $\lim_{x \rightarrow a} (f/g)(x) = \ell/m,$  provided  $m \neq 0.$

*When to apply.* These laws let you compute limits of complicated expressions by breaking them into simpler parts. For instance,  $\lim_{x \rightarrow 2} (3x^2 + x) = 3 \cdot 4 + 2 = 14.$

*Example.*  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2.$  Here direct substitution gives  $0/0$ , so we factor first. The limit laws apply to  $x + 1.$

## Chapter 6. Continuous Functions

**Definition 6.1** (Continuity at a point).  $f$  is **continuous at**  $a$  if  $\lim_{x \rightarrow a} f(x) = f(a).$

Equivalently: for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that  $|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$

**Definition 6.2.**  $f$  is **continuous on**  $[a, b]$  if  $f$  is continuous at every point of  $(a, b),$  right-continuous at  $a,$  and left-continuous at  $b.$

**Theorem 6.3.** If  $f$  and  $g$  are continuous at  $a,$  then  $f + g,$   $f \cdot g,$  and  $f/g$  (when  $g(a) \neq 0$ ) are continuous at  $a.$

**Theorem 6.4.** If  $g$  is continuous at  $a$  and  $f$  is continuous at  $g(a),$  then  $f \circ g$  is continuous at  $a.$

*Example.*  $f(x) = \sin(x^2 + 1)$  is continuous everywhere because  $x^2 + 1$  is a polynomial (continuous), and  $\sin$  is continuous.

## Chapter 7. Three Hard Theorems

These theorems describe *global* properties of continuous functions on closed intervals. They cannot be proved from P1–P12 alone; P13 (the least upper bound property) is essential.

**Theorem 7.1** (Intermediate Value Theorem (IVT)). If  $f$  is continuous on  $[a, b]$  and  $f(a) < 0 < f(b),$  then there exists  $x \in [a, b]$  with  $f(x) = 0.$

*When to apply.* Use the IVT to prove *existence* of solutions to equations. It applies whenever:

- $f$  is continuous on a closed interval, and
- $f$  takes values of opposite sign (or, more generally, a desired value lies between  $f(a)$  and  $f(b).$

It does *not* tell you *where* the root is or how many there are.

*Example.* Show  $x^5 + x = 1$  has a solution. Let  $f(x) = x^5 + x - 1.$  Then  $f(0) = -1 < 0$  and  $f(1) = 1 > 0.$  Since  $f$  is continuous (polynomial), the IVT gives  $x \in (0, 1)$  with  $f(x) = 0.$

**Theorem 7.2** (Boundedness Theorem). If  $f$  is continuous on  $[a, b],$  then  $f$  is bounded above on  $[a, b]:$  there exists  $M$  with  $f(x) \leq M$  for all  $x \in [a, b].$

**Theorem 7.3** (Extreme Value Theorem (Maximum Value Theorem)). If  $f$  is continuous on  $[a, b],$  then  $f$  attains its maximum: there exists  $y \in [a, b]$  with  $f(y) \geq f(x)$  for all  $x \in [a, b].$

*When to apply.* Use this to guarantee that optimization problems have solutions. It requires a *continuous* function on a *closed, bounded* interval—if either hypothesis fails, the conclusion can fail.

*Example.*  $f(x) = 1/x$  on  $(0, 1]$  is continuous but the interval is not closed;  $f$  has no maximum.  $f(x) = x^2$  on  $[-1, 2]$  is continuous on a closed interval, so it attains its max (at  $x = 2$ ,  $f = 4$ ) and its min (at  $x = 0$ ,  $f = 0$ ).

**Corollary 7.4.** *Every positive number has a square root.*

**Corollary 7.5.** *If  $n$  is odd, every polynomial  $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$  has a real root.*

## Chapter 8. Least Upper Bounds

**Definition 8.1** (Upper bound).  $M$  is an **upper bound** for a set  $A \subseteq \mathbb{R}$  if  $x \leq M$  for all  $x \in A$ .  $A$  is **bounded above** if it has an upper bound.

**Definition 8.2** (Least upper bound / Supremum).  $\alpha = \sup A$  if  $\alpha$  is an upper bound for  $A$  and  $\alpha \leq M$  for every upper bound  $M$  of  $A$ .

**Definition 8.3** (Greatest lower bound / Infimum).  $\beta = \inf A$  if  $\beta$  is a lower bound for  $A$  and  $\beta \geq m$  for every lower bound  $m$  of  $A$ .

**Axiom 8.4** (P13 – Least Upper Bound Property (Completeness)). Every nonempty subset of  $\mathbb{R}$  that is bounded above has a least upper bound in  $\mathbb{R}$ .

*Remark.* This is the axiom that distinguishes  $\mathbb{R}$  from  $\mathbb{Q}$ . In  $\mathbb{Q}$ , the set  $\{x \in \mathbb{Q} : x^2 < 2\}$  is bounded above but has no least upper bound in  $\mathbb{Q}$ .

*Example.*  $A = \{1 - 1/n : n \in \mathbb{N}\} = \{0, 1/2, 2/3, 3/4, \dots\}$ . Then  $\sup A = 1$  (which is *not* in  $A$ ), and  $\inf A = 0$  (which *is* in  $A$ ).

## Appendix: Uniform Continuity

**Definition 8.5** (Uniform continuity).  $f$  is **uniformly continuous** on  $A$  if: for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that for *all*  $x, y \in A$ ,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

*Remark.* The key difference from ordinary continuity: the *same*  $\delta$  works for *every* point simultaneously.

**Theorem 8.6.** *If  $f$  is continuous on  $[a, b]$ , then  $f$  is uniformly continuous on  $[a, b]$ .*

*Example.*  $f(x) = 1/x$  is continuous on  $(0, 1)$  but *not* uniformly continuous there, because near 0 you need ever-smaller  $\delta$  for a given  $\varepsilon$ . However,  $f(x) = 1/x$  *is* uniformly continuous on  $[1/2, 1]$  (a closed interval).

## Part III

## Derivatives and Integrals

## Chapter 9. Derivatives

**Definition 9.1** (Derivative).  $f$  is **differentiable at**  $a$  if the limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. The number  $f'(a)$  is called the **derivative of  $f$  at  $a$** .

**Definition 9.2.** The **tangent line** to the graph of  $f$  at  $(a, f(a))$  is the line  $y = f(a) + f'(a)(x - a)$ .

**Definition 9.3** (Higher-order derivatives).  $f'' = (f')'$ ,  $f''' = (f'')'$ , and in general  $f^{(n)} = (f^{(n-1)})'$ .

**Theorem 9.4.** If  $f$  is differentiable at  $a$ , then  $f$  is continuous at  $a$ .

*Remark.* The converse is false:  $f(x) = |x|$  is continuous at 0 but not differentiable there.

*Example.*  $f(x) = x^2$ :  $f'(a) = \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h} = \lim_{h \rightarrow 0} (2a + h) = 2a$ .

## Chapter 10. Differentiation

**Theorem 10.1** (Sum Rule).  $(f + g)'(a) = f'(a) + g'(a)$ .

**Theorem 10.2** (Product Rule).  $(f \cdot g)'(a) = f'(a)g(a) + f(a)g'(a)$ .

*When to apply.* Use the product rule whenever a function is a product of two factors that both depend on  $x$ .

*Example.*  $f(x) = x^2 \sin x$ . Then  $f'(x) = 2x \sin x + x^2 \cos x$ .

**Theorem 10.3** (Quotient Rule). If  $g(a) \neq 0$ , then

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{[g(a)]^2}.$$

**Theorem 10.4** (Chain Rule). If  $g$  is differentiable at  $a$  and  $f$  is differentiable at  $g(a)$ , then  $f \circ g$  is differentiable at  $a$  and

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

*When to apply.* Use the chain rule whenever a function is a *composition*: an “outer” function applied to an “inner” function. Mentally identify the layers and differentiate from outside in.

*Example.*  $f(x) = \sin(x^3)$ . Outer:  $\sin(\cdot)$ , inner:  $x^3$ .  $f'(x) = \cos(x^3) \cdot 3x^2$ .

*Example.*  $f(x) = (1 + x^2)^{100}$ . Then  $f'(x) = 100(1 + x^2)^{99} \cdot 2x$ .

**Theorem 10.5** (Power Rule). For  $f(x) = x^n$  ( $n$  a positive integer),  $f'(x) = nx^{n-1}$ .

## Chapter 11. Significance of the Derivative

**Definition 11.1.** A point  $a$  is a **local maximum** of  $f$  if  $f(a) \geq f(x)$  for all  $x$  in some open interval around  $a$ . Similarly for **local minimum**. A **critical point** is a point where  $f'(a) = 0$ .

**Definition 11.2.**  $f$  is **increasing** on an interval if  $x_1 < x_2 \implies f(x_1) < f(x_2)$ .  $f$  is **decreasing** if  $x_1 < x_2 \implies f(x_1) > f(x_2)$ .

**Theorem 11.3** (Critical Point Theorem). *If  $f$  has a local maximum or minimum at  $a$  and  $f$  is differentiable at  $a$ , then  $f'(a) = 0$ .*

*Remark.* The converse is false:  $f(x) = x^3$  has  $f'(0) = 0$  but 0 is neither a local max nor a local min.

**Theorem 11.4** (Rolle's Theorem). *If  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then there exists  $x \in (a, b)$  with  $f'(x) = 0$ .*

*When to apply.* Rolle's theorem guarantees a horizontal tangent between any two points where the function has the same value. It is the key lemma for the Mean Value Theorem.

*Example.*  $f(x) = x^2 - 4x + 3$  on  $[1, 3]$ . We have  $f(1) = 0 = f(3)$ , so Rolle guarantees  $f'(c) = 0$  for some  $c \in (1, 3)$ . Indeed,  $f'(x) = 2x - 4 = 0$  gives  $c = 2$ .

**Theorem 11.5** (Mean Value Theorem (MVT)). *If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then there exists  $x \in (a, b)$  with*

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

*When to apply.* The MVT is arguably the most important theoretical tool in calculus. It connects the *instantaneous* rate of change to the *average* rate of change. Applications include:

- Proving that  $f' = 0$  everywhere implies  $f$  is constant.
- Proving that  $f' > 0$  everywhere implies  $f$  is increasing.
- Establishing error bounds in approximations.
- Proving L'Hôpital's Rule.

*Example.* If you drive 120 miles in 2 hours, the MVT says there was some instant when your speed was exactly 60 mph.

*Example.* Prove  $|\sin a - \sin b| \leq |a - b|$ . By MVT,  $\sin a - \sin b = \cos(c)(a - b)$  for some  $c$ , so  $|\sin a - \sin b| = |\cos c| |a - b| \leq |a - b|$ .

**Corollary 11.6.** *If  $f'(x) = 0$  for all  $x$  in an interval, then  $f$  is constant on that interval.*

**Corollary 11.7.** *If  $f'(x) = g'(x)$  for all  $x$  in an interval, then  $f = g + c$  for some constant  $c$ .*

**Corollary 11.8.** *If  $f'(x) > 0$  for all  $x$  in an interval, then  $f$  is increasing. If  $f'(x) < 0$  for all  $x$  in an interval, then  $f$  is decreasing.*

**Theorem 11.9** (Second Derivative Test). *If  $f'(a) = 0$ , then:*

- $f''(a) > 0 \implies a$  is a local minimum,
- $f''(a) < 0 \implies a$  is a local maximum.

*If  $f''(a) = 0$ , the test is inconclusive.*

*Example.*  $f(x) = x^3 - 3x$ . Then  $f'(x) = 3x^2 - 3 = 0$  at  $x = \pm 1$ . We have  $f''(x) = 6x$ , so  $f''(1) = 6 > 0$  (local min) and  $f''(-1) = -6 < 0$  (local max).

**Theorem 11.10** (Cauchy Mean Value Theorem). *If  $f, g$  are continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $g'(x) \neq 0$  on  $(a, b)$ , then there exists  $x \in (a, b)$  with*

$$\frac{f'(x)}{g'(x)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

*When to apply.* This generalizes the MVT (take  $g(x) = x$ ) and is used to prove L'Hôpital's Rule.

**Theorem 11.11** (L'Hôpital's Rule). *If  $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$  (or both  $\rightarrow \pm\infty$ ), and  $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$ , then  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$ .*

*When to apply.* Use for indeterminate forms  $0/0$  or  $\infty/\infty$ . Other forms ( $0 \cdot \infty$ ,  $1^\infty$ ,  $0^0$ ,  $\infty^0$ ,  $\infty - \infty$ ) must be rewritten first.

*Example.*  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$

**Theorem 11.12** (Darboux's Theorem). *If  $f$  is differentiable on  $[a, b]$  and  $f'(a) < c < f'(b)$ , then there exists  $x \in (a, b)$  with  $f'(x) = c$ .*

*Remark.* Derivatives satisfy the intermediate value property, even though they need not be continuous!

## Chapter 12. Inverse Functions

**Definition 12.1.**  $f$  is **one-one (injective)** if  $f(x_1) = f(x_2) \implies x_1 = x_2$ . If  $f$  is one-one, its **inverse**  $f^{-1}$  satisfies  $f(f^{-1}(y)) = y$  and  $f^{-1}(f(x)) = x$ .

**Theorem 12.2.** *A continuous one-one function on an interval is either strictly increasing or strictly decreasing.*

**Theorem 12.3.** *If  $f$  is continuous and one-one on an interval, then  $f^{-1}$  is continuous.*

**Theorem 12.4** (Inverse Function Theorem). *If  $f$  is continuous and one-one on an interval, differentiable at  $f^{-1}(b)$  with  $f'(f^{-1}(b)) \neq 0$ , then  $f^{-1}$  is differentiable at  $b$  and*

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

*When to apply.* This is used to find derivatives of all inverse functions:  $\arcsin$ ,  $\arccos$ ,  $\arctan$ ,  $\log$ ,  $x^{1/n}$ , etc.

*Example.* Let  $f(x) = x^3$ , so  $f^{-1}(x) = x^{1/3}$ . Then  $(f^{-1})'(x) = \frac{1}{3(f^{-1}(x))^2} = \frac{1}{3x^{2/3}} = \frac{1}{3}x^{-2/3}$ . This gives us  $(x^{1/3})' = \frac{1}{3}x^{-2/3}$ , consistent with the power rule for rational exponents.

**Corollary 12.5.** *For any rational number  $a$ ,  $(x^a)' = ax^{a-1}$  (where defined).*

## Chapter 13. Integrals

**Definition 13.1** (Partition). A **partition** of  $[a, b]$  is a finite set  $P = \{t_0, t_1, \dots, t_n\}$  with  $a = t_0 < t_1 < \dots < t_n = b$ .

**Definition 13.2** (Upper and lower sums). For a bounded function  $f$  and partition  $P$ :

$$\begin{aligned} m_i &= \inf\{f(x) : t_{i-1} \leq x \leq t_i\}, & M_i &= \sup\{f(x) : t_{i-1} \leq x \leq t_i\}, \\ L(f, P) &= \sum_{i=1}^n m_i(t_i - t_{i-1}), & U(f, P) &= \sum_{i=1}^n M_i(t_i - t_{i-1}). \end{aligned}$$

**Definition 13.3** (Integrable).  $f$  is **integrable on**  $[a, b]$  if  $\sup_P L(f, P) = \inf_P U(f, P)$ . This common value is the **integral**  $\int_a^b f$ .

**Theorem 13.4.** If  $f$  is continuous on  $[a, b]$ , then  $f$  is integrable on  $[a, b]$ .

**Theorem 13.5** (Properties of the integral). If  $f$  and  $g$  are integrable on  $[a, b]$ :

- (1)  $\int_a^b (f + g) = \int_a^b f + \int_a^b g$ ,
- (2)  $\int_a^b cf = c \int_a^b f$ ,
- (3) If  $f \leq g$  on  $[a, b]$ , then  $\int_a^b f \leq \int_a^b g$ ,
- (4)  $\int_a^b f = \int_a^c f + \int_c^b f$  for  $a < c < b$ .

## Chapter 14. The Fundamental Theorem of Calculus

**Theorem 14.1** (First Fundamental Theorem of Calculus). Let  $f$  be integrable on  $[a, b]$  and define  $F(x) = \int_a^x f$ . Then  $F$  is continuous on  $[a, b]$ . Moreover, if  $f$  is continuous at  $c \in [a, b]$ , then  $F$  is differentiable at  $c$  and

$$F'(c) = f(c).$$

*When to apply.* This says that integration and differentiation are inverse processes. Given any continuous  $f$ , the function  $F(x) = \int_a^x f$  is an antiderivative of  $f$ . Use it whenever you need to construct a function with a given derivative.

*Example.*  $F(x) = \int_0^x e^{-t^2} dt$ . Although  $e^{-t^2}$  has no elementary antiderivative, the FTC tells us  $F'(x) = e^{-x^2}$ .

*Example.* Using the chain rule together with the FTC: if  $G(x) = \int_0^{x^2} \sin(t^3) dt$ , then  $G'(x) = \sin(x^6) \cdot 2x$ .

**Theorem 14.2** (Second Fundamental Theorem of Calculus). If  $f$  is integrable on  $[a, b]$  and  $f = g'$  for some function  $g$ , then

$$\int_a^b f = g(b) - g(a).$$

*When to apply.* This is the *evaluation* theorem: to compute  $\int_a^b f$ , find any antiderivative  $g$  and compute  $g(b) - g(a)$ . It turns every derivative formula into an integral formula.

*Example.*  $\int_1^3 x^2 dx = \frac{x^3}{3} \Big|_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}$ .

## Part IV

# Further Topics

## Chapter 15. The Trigonometric Functions

**Definition 15.1.**  $\sin$  and  $\cos$  are defined rigorously (via the area function on the unit circle, or as solutions to  $f'' + f = 0$ ). Key properties:

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1, \\ \sin(a + b) &= \sin a \cos b + \cos a \sin b, \\ \cos(a + b) &= \cos a \cos b - \sin a \sin b.\end{aligned}$$

**Definition 15.2.**  $\pi$  is defined as twice the smallest positive zero of  $\cos$ .

**Theorem 15.3** (Derivatives of trig functions).

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \sec^2 x.$$

**Definition 15.4** (Inverse trig functions).  $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$ ,  $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$ , etc.

**Theorem 15.5** (Derivatives of inverse trig functions).

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\arctan x)' = \frac{1}{1+x^2}.$$

## Chapter 16. $\pi$ is Irrational (Chapter 16\*)

**Theorem 16.1.**  $\pi$  is irrational.

## Chapter 17. Planetary Motion (Chapter 17\*)

Kepler's three laws are derived from Newton's law of gravitation using vector calculus.

## Chapter 18. The Logarithm and Exponential Functions

**Definition 18.1** (Logarithm).  $\log x = \int_1^x \frac{1}{t} dt$ , for  $x > 0$ .

**Theorem 18.2** (Properties of  $\log$ ). (1)  $\log(xy) = \log x + \log y$ ,  
 (2)  $\log(x^n) = n \log x$ ,  
 (3)  $\log$  is increasing with  $\log 1 = 0$ ,  
 (4)  $(\log x)' = 1/x$ .

**Definition 18.3** ( $e$  and  $\exp$ ).  $e$  is the unique number with  $\log e = 1$ . The **exponential function**  $\exp = \log^{-1}$ . Equivalently,  $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n$ .

**Theorem 18.4.**  $(e^x)' = e^x$ , and  $\exp(a + b) = \exp(a) \cdot \exp(b)$ .

**Definition 18.5** (General power). For  $x > 0$  and any real  $a$ ,  $x^a = e^{a \log x}$ .

**Theorem 18.6.**  $(x^a)' = ax^{a-1}$  for all real  $a$  (and  $x > 0$ ).

*When to apply.* This completes the power rule for *all* real exponents, not just rational ones.

**Definition 18.7** (Hyperbolic functions).  $\cosh x = \frac{e^x + e^{-x}}{2}$ ,  $\sinh x = \frac{e^x - e^{-x}}{2}$ ,  $\tanh x = \frac{\sinh x}{\cosh x}$ .

## Chapter 19. Integration in Elementary Terms

**Definition 19.1.** A **primitive** (antiderivative) of  $f$  is a function  $F$  with  $F' = f$ . An **elementary function** is one built from rational functions, trig/inverse trig, log, exp by  $+$ ,  $\times$ ,  $\div$ , and composition.

**Theorem 19.2** (Integration by Parts).

$$\int_a^b f \cdot g' = f \cdot g \Big|_a^b - \int_a^b f' \cdot g.$$

*When to apply.* Use when the integrand is a product, and differentiating one factor while integrating the other simplifies the problem. Classic cases:  $\int x e^x dx$ ,  $\int x \sin x dx$ ,  $\int \ln x dx$ ,  $\int \arctan x dx$ .

*Example.*  $\int x e^x dx$ : let  $f = x$ ,  $g' = e^x$ , so  $f' = 1$ ,  $g = e^x$ .  $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = (x - 1)e^x + C$ .

**Theorem 19.3** (Substitution (Change of Variables)). If  $g$  is differentiable on  $[a, b]$  and  $f$  is continuous on the range of  $g$ :

$$\int_a^b (f \circ g) \cdot g' = \int_{g(a)}^{g(b)} f.$$

*When to apply.* Use when the integrand contains a composite function and (something close to) its derivative. Identify  $u = g(x)$ ,  $du = g'(x) dx$ .

*Example.*  $\int_0^1 2x e^{x^2} dx$ : let  $u = x^2$ ,  $du = 2x dx$ .  $= \int_0^1 e^u du = e - 1$ .

**Theorem 19.4** (Mean Value Theorem for Integrals). If  $f$  is continuous on  $[a, b]$ , there exists  $c \in [a, b]$  with  $\int_a^b f = f(c) \cdot (b - a)$ .

**Theorem 19.5** (Integration of rational functions). Every rational function can be integrated in terms of elementary functions, via partial fraction decomposition.

## Chapter 20. Approximation by Polynomial Functions

**Definition 20.1** (Taylor polynomial). The **Taylor polynomial of degree  $n$**  for  $f$  at  $a$  is

$$P_{n,a}(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

**Definition 20.2** (Remainder).  $R_{n,a}(x) = f(x) - P_{n,a}(x)$ .

**Theorem 20.3** (Taylor's Theorem – Lagrange Remainder). *If  $f^{(n+1)}$  exists on an interval containing  $a$  and  $x$ , then there exists  $c$  between  $a$  and  $x$  with*

$$R_{n,a}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

*When to apply.* Taylor's theorem is the rigorous tool for:

- Approximating functions by polynomials with explicit error bounds.
- Proving convergence of Taylor series.
- Analyzing local behavior of functions near a point.
- Deriving limits involving indeterminate forms.

*Example.*  $e^x$  with  $a = 0$ :  $P_n(x) = \sum_{k=0}^n x^k/k!$ , and  $|R_n(x)| \leq \frac{e^{|x|}}{(n+1)!}|x|^{n+1} \rightarrow 0$  as  $n \rightarrow \infty$ . Therefore  $e^x = \sum_{k=0}^{\infty} x^k/k!$  for all  $x$ .

*Example.*  $\sin x$  with  $a = 0$ :  $P_5(x) = x - x^3/6 + x^5/120$ . The error satisfies  $|R_5(x)| \leq |x|^6/720$ . For  $x = 0.1$ :  $\sin(0.1) \approx 0.0998334$ , with error  $< 1.4 \times 10^{-9}$ .

**Theorem 20.4** (Integral form of the remainder).

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!}(x-t)^n dt.$$

**Theorem 20.5** (Cauchy form of the remainder). *There exists  $c$  between  $a$  and  $x$  with*

$$R_{n,a}(x) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a).$$

## Chapter 21. $e$ is Transcendental (Chapter 21\*)

**Definition 21.1.** A number is **algebraic** if it is a root of a polynomial with integer coefficients. A number is **transcendental** if it is not algebraic.

**Theorem 21.2.**  $e$  is transcendental. (In particular,  $e$  is irrational.)

## Chapter 22. Infinite Sequences

**Definition 22.1** (Convergent sequence).  $\{a_n\}$  **converges to**  $\ell$  (written  $\lim_{n \rightarrow \infty} a_n = \ell$ ) if: for every  $\varepsilon > 0$  there exists  $N$  such that  $n > N \implies |a_n - \ell| < \varepsilon$ .

**Definition 22.2.**  $\{a_n\}$  is **increasing** if  $a_{n+1} > a_n$  for all  $n$ ; **bounded above** if there exists  $M$  with  $a_n \leq M$  for all  $n$ ; and a **Cauchy sequence** if for every  $\varepsilon > 0$  there exists  $N$  with  $|a_m - a_n| < \varepsilon$  whenever  $m, n > N$ .

**Theorem 22.3** (Uniqueness). *A convergent sequence has a unique limit.*

**Theorem 22.4.** *A convergent sequence is bounded.*

**Theorem 22.5** (Limit laws for sequences).  $\lim(a_n + b_n) = \lim a_n + \lim b_n$ ,  $\lim(a_n \cdot b_n) = (\lim a_n)(\lim b_n)$ , etc.

**Theorem 22.6** (Monotone Convergence Theorem). *If  $\{a_n\}$  is nondecreasing and bounded above, then  $\{a_n\}$  converges (to  $\sup\{a_n\}$ ).*

*When to apply.* Use this whenever you can show a sequence is both monotone and bounded. Classic examples: recursive sequences like  $a_{n+1} = \sqrt{2 + a_n}$ .

*Example.*  $a_1 = 1$ ,  $a_{n+1} = \sqrt{2 + a_n}$ . One shows by induction that  $a_n < 2$  (bounded) and  $a_{n+1} > a_n$  (increasing). By the MCT,  $\ell = \lim a_n$  exists, and  $\ell = \sqrt{2 + \ell}$  gives  $\ell = 2$ .

**Theorem 22.7** (Bolzano–Weierstrass Theorem). *Every bounded sequence has a convergent subsequence.*

*When to apply.* This is a fundamental compactness result. Use it when you know a sequence is bounded and need to extract a convergent subsequence, even if the original sequence itself may not converge.

**Theorem 22.8** (Cauchy Criterion). *A sequence converges if and only if it is a Cauchy sequence.*

*When to apply.* Useful when you want to prove convergence without knowing the limit. You only need to show terms get arbitrarily close to each other.

## Chapter 23. Infinite Series

**Definition 23.1.** The **infinite series**  $\sum_{n=1}^{\infty} a_n$  is defined as  $\lim_{N \rightarrow \infty} s_N$  where  $s_N = \sum_{n=1}^N a_n$  (partial sums). If this limit exists, the series **converges**; otherwise it **diverges**.

**Definition 23.2.**  $\sum a_n$  is **absolutely convergent** if  $\sum |a_n|$  converges. It is **conditionally convergent** if it converges but  $\sum |a_n|$  diverges.

**Theorem 23.3** (Vanishing Condition / Divergence Test). *If  $\sum a_n$  converges, then  $a_n \rightarrow 0$ .*

*Remark.* The converse is false:  $\sum 1/n$  diverges even though  $1/n \rightarrow 0$ .

**Theorem 23.4** (Geometric Series).  $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$  for  $|x| < 1$ ; *diverges for  $|x| \geq 1$ .*

**Theorem 23.5** (Comparison Test). *If  $0 \leq a_n \leq b_n$  for all  $n$  and  $\sum b_n$  converges, then  $\sum a_n$  converges. If  $\sum a_n$  diverges, then  $\sum b_n$  diverges.*

*When to apply.* Compare a complicated series to a simpler one (geometric,  $p$ -series, etc.) to determine convergence.

*Example.*  $\sum \frac{1}{n^2+1}$ : since  $\frac{1}{n^2+1} \leq \frac{1}{n^2}$  and  $\sum 1/n^2$  converges ( $p$ -series,  $p = 2 > 1$ ), the series converges.

**Theorem 23.6** (Ratio Test). *Let  $r = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$  (if it exists). Then:  $r < 1 \implies$  absolutely convergent;  $r > 1 \implies$  divergent;  $r = 1 \implies$  inconclusive.*

*When to apply.* Best for series involving factorials and exponentials.

*Example.*  $\sum \frac{n!}{n^n}$ :  $\frac{a_{n+1}}{a_n} = \frac{(n+1)! n^n}{(n+1)^{n+1} n!} = \left( \frac{n}{n+1} \right)^n \rightarrow e^{-1} < 1$ . Converges.

**Theorem 23.7** (Root Test). *Let  $r = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$ . Then:  $r < 1 \implies$  absolutely convergent;  $r > 1 \implies$  divergent;  $r = 1 \implies$  inconclusive.*

*When to apply.* Best for series whose general term has the form  $(c_n)^n$ .

**Theorem 23.8** (Integral Test). *If  $f$  is positive, decreasing, and continuous on  $[1, \infty)$ , then  $\sum_{n=1}^{\infty} f(n)$  converges  $\iff \int_1^{\infty} f$  converges.*

*Example.*  $\sum 1/n^p$ : let  $f(x) = x^{-p}$ . Then  $\int_1^\infty x^{-p} dx$  converges iff  $p > 1$ . So  $\sum 1/n^p$  converges iff  $p > 1$ .

**Theorem 23.9** (Alternating Series Test (Leibniz)). *If  $a_n \geq 0$ ,  $\{a_n\}$  is decreasing, and  $a_n \rightarrow 0$ , then  $\sum (-1)^n a_n$  converges.*

*Example.*  $\sum_{n=1}^\infty \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2$ . This converges conditionally (not absolutely, since  $\sum 1/n$  diverges).

**Theorem 23.10** (Absolute convergence implies convergence). *If  $\sum |a_n|$  converges, then  $\sum a_n$  converges.*

**Theorem 23.11** (Rearrangement). *If  $\sum a_n$  is conditionally convergent, then for any real number  $L$  (or  $\pm\infty$ ) there exists a rearrangement of the series that converges to  $L$  (or diverges).*

## Chapter 24. Uniform Convergence and Power Series

**Definition 24.1** (Uniform convergence of functions).  $\{f_n\}$  **converges uniformly** to  $f$  on  $A$  if: for every  $\varepsilon > 0$  there exists  $N$  such that for all  $n > N$  and all  $x \in A$ ,  $|f(x) - f_n(x)| < \varepsilon$ .

**Theorem 24.2** (Uniform limit of continuous functions is continuous). *If each  $f_n$  is continuous on  $[a, b]$  and  $f_n \rightarrow f$  uniformly, then  $f$  is continuous on  $[a, b]$ .*

*Remark.* This can fail for pointwise convergence:  $f_n(x) = x^n$  on  $[0, 1]$  converges pointwise to a discontinuous function.

**Theorem 24.3** (Interchanging limit and integral). *If  $f_n \rightarrow f$  uniformly on  $[a, b]$  and each  $f_n$  is integrable, then  $\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n$ .*

**Theorem 24.4** (Interchanging limit and derivative). *If  $\{f_n\}$  converges pointwise to  $f$ , each  $f_n$  has an integrable derivative, and  $\{f'_n\}$  converges uniformly to  $g$ , then  $f' = g$ , i.e.,  $\lim f'_n = (\lim f_n)'$ .*

**Theorem 24.5** (Weierstrass  $M$ -test). *If  $|f_n(x)| \leq M_n$  for all  $x \in A$  and  $\sum M_n$  converges, then  $\sum f_n$  converges uniformly (and absolutely) on  $A$ .*

*When to apply.* This is the standard tool for proving uniform convergence of series of functions. Choose constants  $M_n$  that dominate  $|f_n(x)|$  uniformly and form a convergent numerical series.

*Example.*  $\sum \frac{\sin(nx)}{n^2}$  converges uniformly on  $\mathbb{R}$ : take  $M_n = 1/n^2$ . Since  $\sum 1/n^2$  converges, the  $M$ -test applies.

**Theorem 24.6** (Weierstrass: continuous, nowhere-differentiable function).  $f(x) = \sum_{n=1}^\infty \frac{\{10^n x\}}{10^n}$  is continuous everywhere but differentiable nowhere.

**Definition 24.7** (Power series). A **power series centered at  $a$**  is  $\sum_{n=0}^\infty c_n(x-a)^n$ .

**Definition 24.8** (Radius of convergence). The **radius of convergence**  $R$  of  $\sum c_n x^n$  is the number such that the series converges absolutely for  $|x| < R$  and diverges for  $|x| > R$ .  $R = 1/\limsup |c_n|^{1/n}$ .

**Theorem 24.9** (Differentiation of power series). *If  $\sum a_n x^n$  has radius of convergence  $R > 0$ , then  $f(x) = \sum a_n x^n$  is differentiable for  $|x| < R$  and*

$$f'(x) = \sum_{n=1}^\infty n a_n x^{n-1} \quad \text{for } |x| < R.$$

*The differentiated series has the same radius of convergence.*

*When to apply.* Power series can be differentiated (and integrated) term by term inside the radius of convergence. This is enormously useful for finding Taylor series of new functions from known ones.

**Theorem 24.10** (Abel's Theorem). *If  $\sum_{n=0}^{\infty} a_n$  converges, then  $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n$ .*

*Example.* Since  $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} x^n / n$  for  $|x| < 1$ , Abel's theorem shows  $\ln 2 = \sum_{n=1}^{\infty} (-1)^{n+1} / n$ .

**Definition 24.11** (Taylor series). The **Taylor series** for  $f$  at  $a$  is  $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$ .

## Chapter 25. Complex Numbers

**Definition 25.1.** A **complex number** is  $z = a + bi$  where  $i^2 = -1$ . The **modulus** is  $|z| = \sqrt{a^2 + b^2}$ , the **conjugate** is  $\bar{z} = a - bi$ .

**Theorem 25.2** (De Moivre's Theorem).  $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ .

**Theorem 25.3** (Complex  $n$ th roots). *Every nonzero complex number has exactly  $n$  distinct  $n$ th roots:*

$$z^{1/n} = r^{1/n} \left( \cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right), \quad k = 0, 1, \dots, n-1.$$

**Theorem 25.4** (Fundamental Theorem of Algebra). *Every polynomial of degree  $\geq 1$  with complex coefficients has at least one root in  $\mathbb{C}$ .*

*When to apply.* Every polynomial of degree  $n$  factors completely into  $n$  linear factors over  $\mathbb{C}$ .

## Chapter 26. Complex Functions

**Definition 26.1.** A complex function  $f$  is **differentiable at  $z$**  if  $\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$  exists (with  $h$  complex).

**Definition 26.2.** An **entire function** is a complex function differentiable everywhere in  $\mathbb{C}$ .

## Chapter 27. Complex Power Series

**Definition 27.1.**  $\sum a_n (z-a)^n$  is a **complex power series**. Its **circle of convergence** has radius  $R = 1 / \limsup |a_n|^{1/n}$  (Cauchy-Hadamard formula).

**Theorem 27.2** (Euler's Formula).  $e^{i\theta} = \cos \theta + i \sin \theta$ .

*When to apply.* This connects exponential and trigonometric functions, and gives the polar form  $z = re^{i\theta}$ . It also implies  $e^{i\pi} + 1 = 0$ .

## Part V

# Epilogue

## Chapter 28. Fields

**Definition 28.1.** A **field** is a set  $F$  with operations  $+$  and  $\cdot$  satisfying axioms analogous to P1–P9.

**Definition 28.2.** An **ordered field** is a field  $F$  with a subset  $P$  (“positive elements”) satisfying P10–P12.

**Definition 28.3.** A **complete ordered field** is an ordered field in which every nonempty set bounded above has a least upper bound.

## Chapter 29. Construction of the Real Numbers

**Definition 29.1** (Dedekind cut). A **real number** (Dedekind cut) is a subset  $\alpha \subset \mathbb{Q}$  such that:

- (i)  $\alpha \neq \emptyset$  and  $\alpha \neq \mathbb{Q}$ ,
- (ii) if  $x \in \alpha$  and  $y < x$ , then  $y \in \alpha$  (downward closed),
- (iii)  $\alpha$  has no greatest element.

**Theorem 29.2.** *The set of Dedekind cuts forms a complete ordered field.*

## Chapter 30. Uniqueness of the Real Numbers

**Theorem 30.1** (Uniqueness). *Any two complete ordered fields are isomorphic.*

*Remark.* This means that  $\mathbb{R}$  is the *unique* complete ordered field (up to isomorphism). Whether we construct it via Dedekind cuts, Cauchy sequences, or any other method, we always get the same mathematical structure.